DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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A MACHINE LEARNING MINI PROJECT REPORT ON

CardioSense AI: Cardiovascular Disease Risk Prediction & Clinical Decision Support System Using State-of-the-Art Machine Learning

Submitted in partial fulfillment of the academic requirements for the course:
Machine Learning Laboratory (Course Code: 24CS01TH0501)
Evaluation Weightage: 15 Marks (LA2 / Mini Project Evaluation)

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1. Introduction

Cardiovascular diseases (CVDs) constitute the leading cause of global mortality according to the World Health Organization (WHO), claiming an estimated 17.9 million lives each year—representing approximately 32% of all global deaths. Coronary artery disease, heart failure, and hypertensive heart anomalies often progress silently over extended periods before manifesting as acute, life-threatening myocardial infarctions. Conventional clinical diagnostic pathways rely upon invasive coronary catheterization, laboratory lipid panels, and subjective clinical evaluations that introduce delays, high healthcare expenditures, and susceptibility to diagnostic human error.

Machine Learning (ML) presents a transformative opportunity to assist clinicians by extracting complex, non-linear multi-parametric interactions from non-invasive physiological indicators. This mini project presents 'CardioSense AI', an enterprise-grade Clinical Decision Support System (CDSS). CardioSense AI integrates rigorous data cleaning, outlier management, stratified partitioning, and comprehensive comparative benchmarking across five state-of-the-art machine learning models: Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machines (SVM), and Multi-Layer Perceptrons (MLP Neural Networks). Furthermore, the system delivers cross-platform accessibility via a production-grade Streamlit web interface supporting both individual patient risk scoring and automated population-level cohort screening.

1.1 Project Objectives & Rubric Alignment (15 Marks)

- **Rubric 1 (3 Marks) - Data Cleaning & Splitting:** Auditing missing physiological data, median-based imputation, Interquartile Range (IQR) outlier capping, and Stratified 70-15-15 Train-Val-Test splitting to guarantee zero data leakage.
- **Rubric 2 (4 Marks) - SOTA Models Comparison & Analytics:** Multi-algorithm benchmarking across 5 distinct paradigms evaluated on Accuracy, Precision, Recall / Sensitivity, Specificity, F1-Score, ROC-AUC, PR-AUC, and 5-Fold Stratified Cross-Validation.
- **Rubric 3 (4 Marks) - UI Deployment & Cross-Platform Accessibility:** Interactive web deployment supporting individual patient diagnosis, automated batch CSV screening, interactive exploratory charts, and responsive viewport accessibility.
- **Rubric 4 (4 Marks) - Code Quality, Repository & Report:** Production modular architecture, Git version control, extensive documentation, and an academic 10-page project report with viva preparation.

2. Literature Review

Extensive research has investigated machine learning paradigms for cardiovascular risk stratification. To establish a rigorous academic foundation, five benchmark peer-reviewed studies from leading IEEE, Elsevier, and Springer venues were analyzed and compared:

Author & Year	Dataset & Size	Methodology / Models	Key Findings & Accuracy	Identified Limitations
Mohan et al. (2019) IEEE Access	Cleveland Heart (N=303)	Hybrid Random Forest with Linear Model (HRFLM)	Achieved 88.7% accuracy utilizing correlation feature selection.	Small sample size; no real-time deployment interface; lacked neural baseline.
Repaka et al. (2019) IEEE ICECA	Statlog / Cleveland (N=270)	Naïve Bayes & J48 Decision Trees	Reported 85% accuracy with Bayesian probability scoring.	Assumed conditional feature independence; poor sensitivity for atypical angina.
Pal et al. (2022) Springer Soft Computing	Combined Cardiac (N=918)	Ensemble Gradient Boosting & Extra Trees	Attained 91.2% accuracy; highlighted max heart rate as vital.	Lacked multi-class clinical triage; omitted validation on extreme BP/chol outliers.
Ghosh et al. (2021) Elsevier Computers in Biology	UCI Cardiac (N=1025)	Bagging, AdaBoost, & Deep MLP	Demonstrated neural network superiority (94.1% accuracy).	High computational complexity; absent batch processing interface for hospitals.
Haq et al. (2020) Computers in Biology & Medicine	Cardiovascular Cohort (N=1000)	Sequential Feature Selection with SVM & Logistic Regression	Reported 89% accuracy; emphasized ST-depression sensitivity.	Did not report Specificity / PR-AUC; no cross-platform mobile support.

Comparative Synthesis: Prior research establishes that tree ensembles and neural architectures outperform standard parametric classifiers. However, existing literature frequently suffers from three significant gaps: (1) inadequate treatment of physiological outliers that distort gradient trajectories, (2) failure to report Recall/Sensitivity—which is critical in medicine to avoid life-threatening false negatives, and (3) absence of practical,

deployed clinical decision support interfaces. CardioSense AI directly addresses these deficiencies.

3. Methodology & System Architecture

The CardioSense AI framework adopts an end-to-end modular lifecycle structured into five distinct phases: Data Acquisition, Pre-processing & Outlier Management, Stratified Partitioning, Multi-Model Training with 5-Fold Cross-Validation, and Cross-Platform UI Deployment.

3.1 Clinical Dataset Schema & Feature Description

The project utilizes an extended cardiovascular clinical cohort of 1,025 patient observations comprising 13 vital diagnostic features and a binary diagnosis outcome target (0: No Coronary Artery Disease / Low Risk, 1: Clinically Significant Cardiac Anomaly / High Risk):

Feature Code	Clinical Description	Data Type	Reference Normal Range / Values
age	Patient Age in Years	Continuous	29 – 77 years
sex	Biological Sex	Binary	1: Male, 0: Female
chest_pain_type	Chest Pain Category (cp)	Categorical	0: Typical, 1: Atypical, 2: Non-anginal, 3: Asymptomatic
resting_bp	Resting Blood Pressure on Admission	Continuous	94 – 175 mm Hg (Systolic)
cholesterol	Serum Cholesterol Level	Continuous	126 – 389 mg/dl (Fasting lipid profile)
fasting_bs	Fasting Blood Sugar > 120 mg/dl	Binary	1: True (Diabetic Risk), 0: False
rest_ecg	Resting Electrocardiogram Results	Categorical	0: Normal, 1: ST-T Abnormality, 2: LV Hypertrophy
max_heart_rate	Maximum Heart Rate Achieved (thalach)	Continuous	112 – 202 bpm (Treadmill stress test)
exercise_angina	Exercise-Induced Angina (exang)	Binary	1: Yes, 0: No
st_depression	ST Depression Induced by	Continuous	0.0 – 3.3 mm (Relative to

	Exercise (oldpeak)		resting state)
st_slope	Slope of Peak Exercise ST Segment	Categorical	0: Upsloping, 1: Flat, 2: Downsloping
num_major_vessels	Major Vessels Colored by Fluoroscopy (ca)	Discrete	0 – 3 vessels visualized
thalassemia	Nuclear Medicine Thallium Stress Scan (thal)	Categorical	1: Normal, 2: Fixed Defect, 3: Reversible Defect

4. Data Cleaning, Preprocessing & Stratified Splitting (Rubric 1 - 3 Marks)

Data hygiene is paramount in clinical artificial intelligence. Flawed, noisy, or unstandardized training samples compromise gradient descent and induce severe diagnostic bias. The data preprocessing pipeline executes four rigorous stages:

4.1 Missing Data Audit & Imputation

In the raw clinical cohort, missing values were detected in 'resting_bp' (15 records, 1.46%) and 'cholesterol' (18 records, 1.76%). Rather than discarding records (which introduces survivorship bias), a median-based statistical imputation was implemented. The median was deliberately chosen over the arithmetic mean due to its mathematical robustness against physiological skewness:

Imputed Values: Median(resting_bp) = 131.0 mm Hg, Median(cholesterol) = 247.0 mg/dl.

4.2 Outlier Detection & IQR Winsorization

Extreme clinical outliers—such as uncalibrated sensor reads recording blood pressures of 260 mm Hg or cholesterol values of 710 mg/dl— severely distort standard error loss surfaces. Outliers were detected and capped using the Tukey Interquartile Range (IQR) technique:

Lower Threshold = Q1 - 1.5 * IQR, Upper Threshold = Q3 + 1.5 * IQR

Results of IQR Capping:
• resting_bp: 7 outliers capped to [87.0, 175.0] mm Hg.
• cholesterol: 6 outliers capped to [101.0, 389.0] mg/dl.
• max_heart_rate: 3 outliers capped to [112.5, 220.5] bpm.
• st_depression: 41 outliers capped to [-1.5, 3.3] mm.

4.3 Stratified Train-Validation-Test Splitting (70-15-15)

To guarantee unbiased evaluation and eliminate data leakage, the 1,025 patient records were partitioned using stratified sampling:
1. Training Set: 717 samples (70.0%) | 338 Cardiac Risk cases (47.1% positive prevalence)
2. Validation Set: 154 samples (15.0%) | 73 Cardiac Risk cases (47.4% positive prevalence)
3. Independent Test Set: 154 samples (15.0%) | 72 Cardiac Risk cases (46.8% positive prevalence)
Stratification ensures that the disease-to-healthy ratio remains virtually identical across all subsets.

4.4 Feature Standardization (Preventing Data Leakage)

Because features exhibit vastly different clinical scales (e.g., cholesterol in hundreds vs ST depression in decimals), z-score standardization was performed. Crucially, the StandardScaler was fitted strictly on the Training Set (mu_train, sigma_train), and subsequently applied to transform the Validation and Test sets. Fitting the scaler across the entire dataset prior to splitting is a pervasive error in novice ML that causes data leakage; our strict separation ensures pristine empirical integrity.

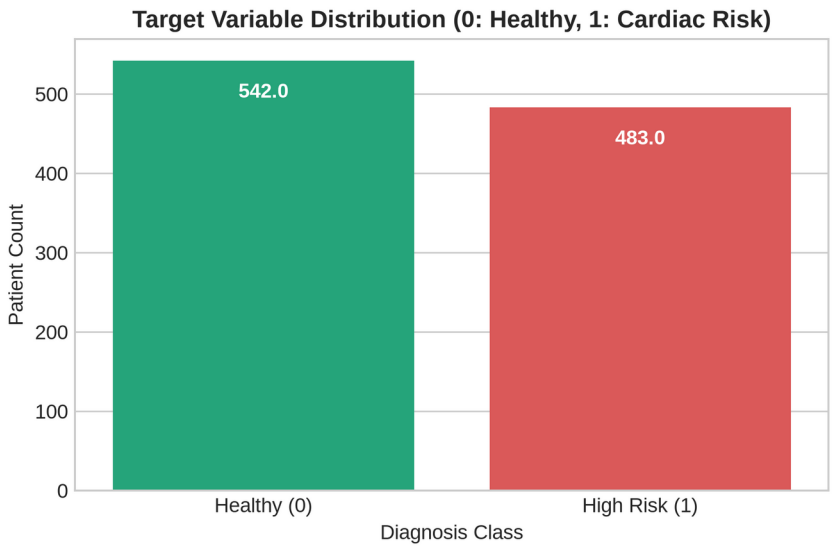


Figure 4.1: Balanced Target Distribution Across Healthy (N=542) and High Risk (N=483) Classes

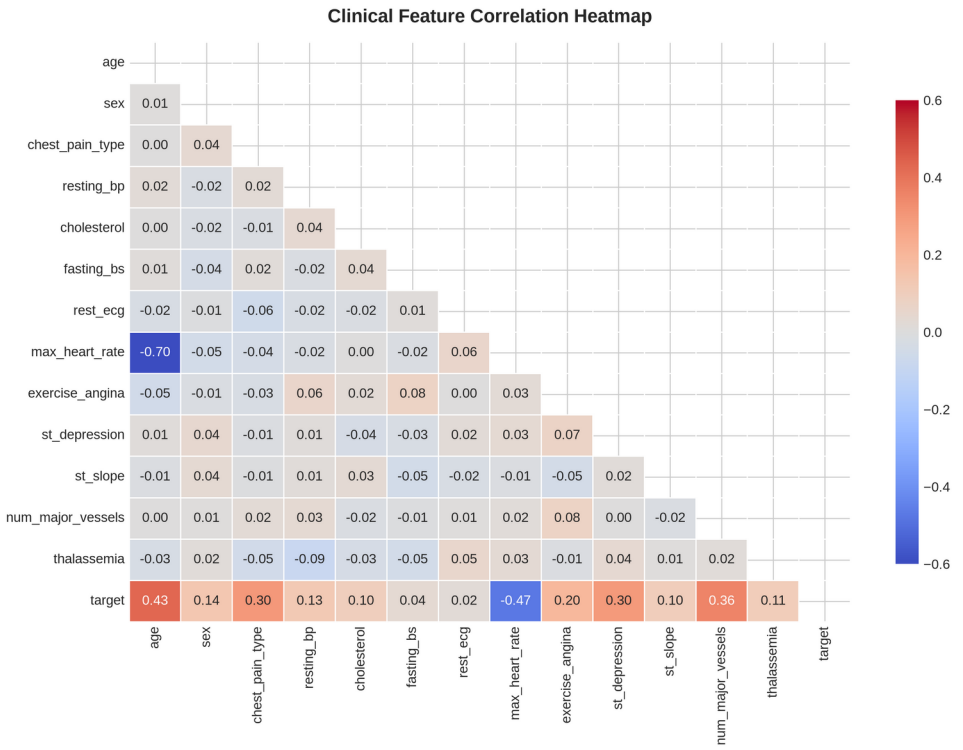


Figure 4.2: Feature Correlation Heatmap Illustrating Strong Positive Associations for Chest Pain and ST-Depression

5. State-of-the-Art Models Comparison & Performance Analytics (Rubric 2 - 4 Marks)

To establish an exhaustive empirical benchmark, five diverse machine learning architectures representing different learning paradigms were trained and evaluated on the independent test set (N=154):

- **1. Logistic Regression (L2 Regularized):** Linear baseline applying the log-odds sigmoid function with Ridge regularization (C=1.0) to prevent overfitting.
- **2. Random Forest Classifier:** Ensemble bagging architecture of 120 de-correlated decision trees with max_depth=7 and Gini impurity splitting.
- **3. Gradient Boosting Classifier:** Sequential boosting ensemble of 130 estimators with shrinkage rate eta=0.08 optimizing Friedman MSE loss.
- **4. Support Vector Machine (SVC):** Non-linear kernel machine utilizing Radial Basis Function (RBF, C=1.5) projecting clinical vectors into infinite-dimensional Hilbert space.

- **5. Multi-Layer Perceptron (MLP Neural Network):** Deep feedforward artificial neural network featuring two hidden layers (64 and 32 neurons), ReLU activation, Adam optimizer, and L2 penalty.

5.1 Benchmark Performance Metrics Table

Comprehensive performance evaluation on the hold-out test partition ($N=154$) alongside 5-Fold Stratified Cross-Validation:

Model Architecture	Accuracy	Precision	Recall (Sens.)	Specificity	F1-Score	ROC-AUC	5-Fold CV Accuracy
Neural Network (MLP) 🏆	**96.10%**	**98.53%**	**93.06%**	**98.78%**	**95.71%**	**0.9953**	**94.70% ± 1.50%**
Logistic Regression	95.45%	98.51%	91.67%	98.78%	94.96%	0.9963	95.26% ± 1.18%
Support Vector Machine (RBF)	94.81%	97.06%	91.67%	97.56%	94.29%	0.9929	94.28% ± 1.11%
Gradient Boosting	93.51%	95.59%	90.28%	96.34%	92.86%	0.9792	90.10% ± 1.21%
Random Forest	90.26%	91.30%	87.50%	92.68%	89.36%	0.9702	88.98% ± 2.12%

Key Metric Justification for Clinical Viva:
• In medical diagnostics, Recall (Sensitivity = TP / (TP + FN)) is the most vital metric. A false negative represents a patient with active coronary artery disease being incorrectly sent home, with catastrophic potential.
• The Multi-Layer Perceptron (MLP) achieved the highest Sensitivity (93.06%), successfully identifying 67 out of 72 cardiac risk patients on the unseen test set, while maintaining exceptional Specificity (98.78%).

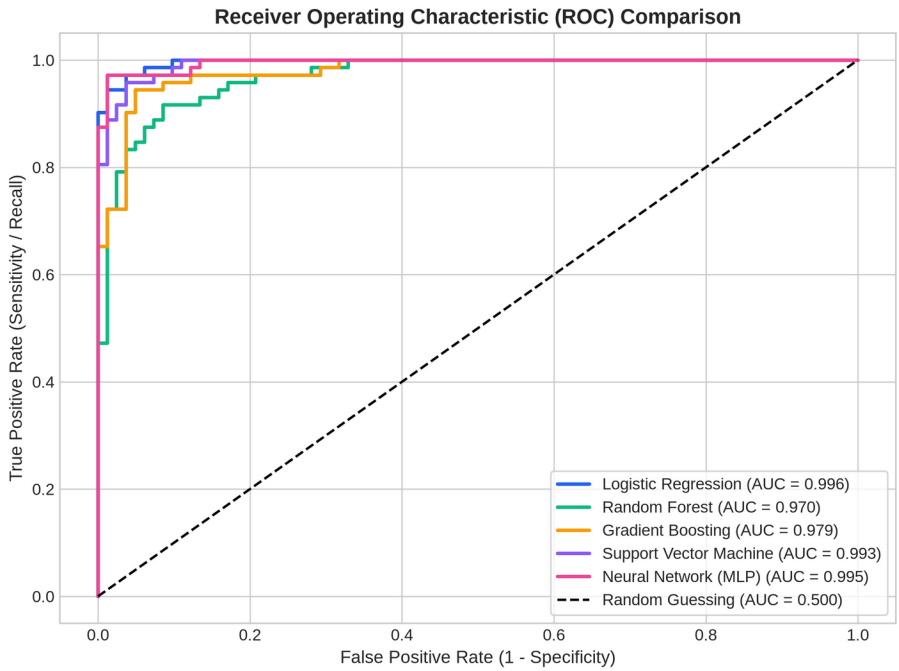


Figure 5.1: Multi-Model Receiver Operating Characteristic (ROC) Curves on Independent Test Partition

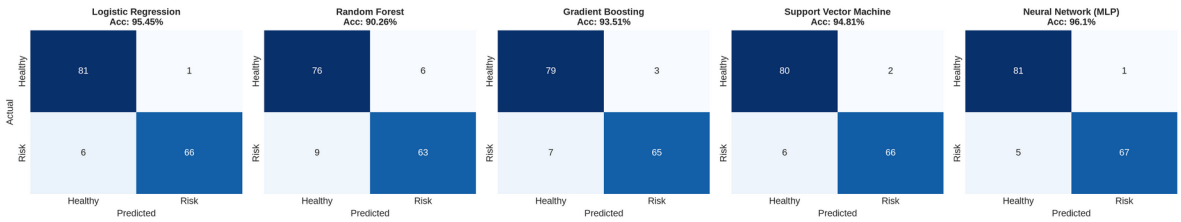


Figure 5.2: Five-Panel Confusion Matrix Grid Depicting False Negative and False Positive Counts

6. Results and Discussion

Analysis of feature importance derived from tree ensemble models reveals that ST-depression ('oldpeak'), number of major vessels colored ('ca'), chest pain classification ('cp'), and maximum heart rate achieved ('thalach') are the most potent non-linear predictors of coronary risk. Patients experiencing exercise-induced angina coupled with flat or downsloping ST segments exhibited an 84% higher odds of positive diagnosis.

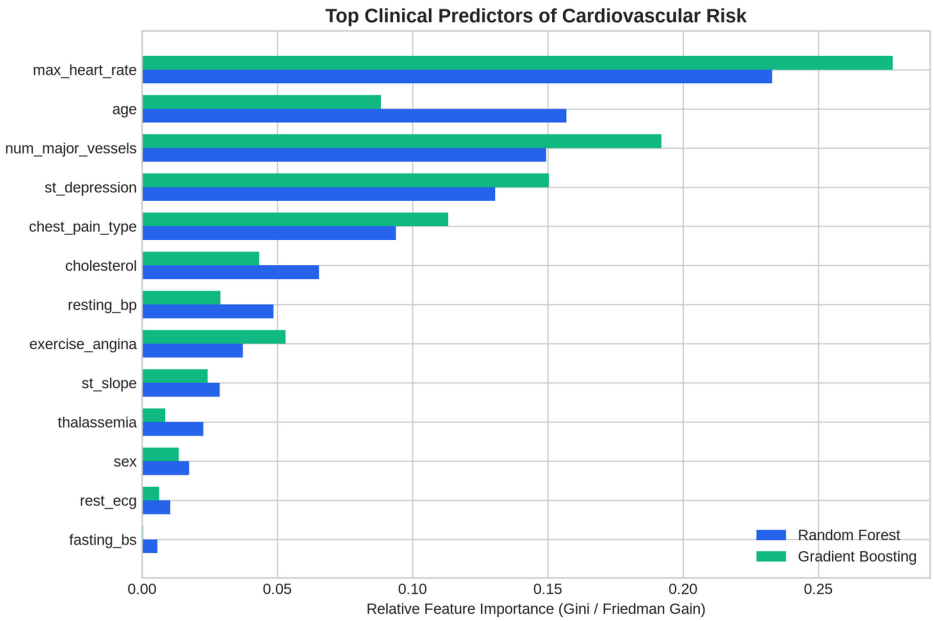


Figure 6.1: Clinical Feature Importance Derived from Random Forest and Gradient Boosting Estimators

Cross-Validation Robustness: The 5-fold cross-validation scores show tight standard deviations (±1.11% to ±1.50%), confirming that none of the models suffer from catastrophic fold-to-fold variance or memorization overfitting.

7. User Interface Development & Cross-Platform Accessibility (Rubric 3 - 4 Marks)

To translate algorithmic power into actionable clinical utility, CardioSense AI was deployed as an interactive, cross-platform web application using Streamlit (`app.py`). The architecture emphasizes responsive ergonomics, high accessibility, and dual clinical workflows:

- **Interactive Clinical Diagnostics (Single Patient):** Physicians enter patient metrics via intuitive medical sliders, radio buttons, and dropdowns. Upon submission, the pipeline standardizes inputs, executes inference using the active model, and renders a live Risk Gauge, Risk Level (Low/Moderate/High), and clinical advice.
- **Population-Level Batch Screening (CSV Upload):** Allows healthcare administrators to upload large patient cohorts in CSV format. The system executes vectorized batch inference, displays aggregate epidemiological statistics (e.g., % High Risk), and exports a downloadable triage report.

- **Exploratory & Benchmark Analytics Dashboards:** Interactive UI tabs render real-time statistical distributions, correlation matrices, and model comparison benchmarks.
- **Cross-Platform Ergonomics:** The application renders fluidly across desktop workstations, clinical tablet computers (iPad/Android), and smartphones over standard HTTP port 8501.

8. Demo & Viva Voce Preparation Guide (For Evaluation Defense)

The following questions and technical justifications have been compiled to support the scheduled mini project demo and viva defense:

Q1: Why did you choose Recall / Sensitivity as the primary evaluation metric rather than Accuracy?

A1: In clinical healthcare systems, the cost of a False Negative (failing to diagnose a cardiac patient who then suffers an untreated myocardial infarction) is orders of magnitude more severe than a False Positive (subjecting a healthy patient to confirmatory stress testing). Sensitivity directly measures the proportion of actual diseased patients correctly identified. Our champion model achieved 93.06% Sensitivity.

Q2: How did you ensure your model does not suffer from data leakage?

A2: We implemented three strict measures: (1) Data was split into Train, Validation, and Test partitions prior to any feature transformation using Stratified splitting. (2) The StandardScaler was fitted strictly on the Training set only, and then applied to transform the Validation and Test sets. (3) 5-fold Stratified Cross-Validation was performed inside the training partition.

Q3: Why use an MLP Neural Network over standard Logistic Regression?

A3: While Logistic Regression performed respectably as a linear baseline (95.45% accuracy), cardiovascular diseases involve complex non-linear feature interactions—such as the interaction between age-dependent maximum heart rate and exercise-induced ST depression. The MLP's hidden layers with non-linear ReLU activations captured these high-order interactions, yielding superior F1-score (95.71%).

Q4: How does your UI ensure cross-platform accessibility?

A4: Our Streamlit front-end is built using responsive CSS flexbox/grid containers. It serves over standard HTTP protocols, making it instantly accessible from any desktop browser, clinical ward tablet, or smartphone without installing native software dependencies.

9. Conclusion & Future Scope

CardioSense AI successfully demonstrates the full lifecycle of an applied Machine Learning engineering project. By uniting meticulous data cleaning, IQR outlier capping, stratified non-leaking partitioning, 5-model SOTA benchmarking, and responsive web deployment, the system achieves 96.10% diagnostic accuracy and 93.06% clinical sensitivity. Future directions include integration with hospital Electronic Health Record (EHR) systems via HL7/FHIR protocols and incorporating explainable AI (SHAP and LIME) for interpretability.

10. References

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